

MATH 521

Real Analysis

Final Practice Book

Write-On Workbook

University of Wisconsin–Madison

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Step 1 build: Homework 9

Name: _____

Date: _____

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How to Use This Workbook

This workbook is for active proof practice. For each problem, work through the diagnosis, recipe card, self-explanation, skeleton, and full proof attempt before reading the reference solution.

- Pass 1** Diagnose the topic, task, relevant definitions, and likely proof method.
- Pass 2** Convert the problem into hypotheses, conclusion, tools, and a proof plan.
- Pass 3** Answer targeted questions that expose the proof's logic.
- Pass 4** Reconstruct the proof from a skeleton without reading the full solution.
- Pass 5** Write a complete proof from scratch in the blank writing space.

Only after Pass 5 should you compare your work to the official or reference solution.

Homework 9

Problem 1

Problem Statement

Homework 9, Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

- (a) $A = [0, 1)$
- (b) $A = \mathbb{Z}$
- (c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. For each set, which points are obvious members of the closure, and which boundary points require a ball argument?

2. How do you prove that a point is not in $\overline{\mathbb{Z}}$? What radius separates it from all integers?

3. What is the difference between proving $a \in A$ is isolated and proving $a \in A$ is a limit point?

4. Where is density of $\mathbb{Q}$ used in part (c), and why is it not enough just to say “there are rationals nearby”?

5. How does the identity $\overline{A} = (\text{isolated points}) \cup A'$ finish the limit-point computations?

Part (a): $A = [0, 1)$

Closure: show $[0, 1) \subseteq \overline{A} \subseteq [0, 1]$, then prove $1 \in \overline{A}$ by choosing a point of A in $(1 - r, 1 + r)$:

Isolated points: fix $a \in A$. Given $r > 0$, choose a second point of A inside $(a - r, a + r)$:

Limit points: use the closure/isolated-point relationship to conclude:

Part (b): $A = \mathbb{Z}$

Closure: for $x \notin \mathbb{Z}$, choose $n \in \mathbb{Z}$ with $n < x < n + 1$ and radius $r =$

Isolated points and limit points: choose the ball around $n \in \mathbb{Z}$:

Part (c): $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Closure: prove 0, points in $(0, 1]$, and no points outside $[0, 1]$ behave correctly:

Isolated points and limit points: use density of $\mathbb{Q}$ to find a second point:

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

- (a) Closure: $\overline{A} = [0, 1]$. Since $\overline{A}$ is the smallest closed set containing A , and $[0, 1]$ is one such closed set, we must have

$$[0, 1) \subseteq \overline{A} \subseteq [0, 1].$$

Also $1 \in \overline{A}$: given $r > 0$,

$$(1 - r, 1 + r) \cap [0, 1] \neq \emptyset,$$

since $\max\{1 - r/2, 0\} \in (1 - r, 1 + r) \cap [0, 1]$.

Isolated points: $\emptyset$. Fix $a \in A$. Given $r > 0$, the point

$$\min\left\{a + \frac{r}{2}, \frac{a + 1}{2}\right\}$$

is in $(a - r, a + r) \cap A$ and is not a .

Limit points: $A' = [0, 1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and there are no isolated points.

- (b) Closure: $\overline{A} = \mathbb{Z}$. We automatically have $\mathbb{Z} \subseteq \overline{A}$. Fix $x \in \mathbb{Z}^c$. Let $n \in \mathbb{Z}$ satisfy $n < x < n + 1$. Then

$$r = \min\{x - n, n + 1 - x\}$$

is positive and $(x - r, x + r) \cap \mathbb{Z} = \emptyset$. Thus $x \notin \overline{A}$.

Isolated points: $\mathbb{Z}$. For $n \in \mathbb{Z}$,

$$(n - 1, n + 1) \cap \mathbb{Z} = \{n\}.$$

Limit points: $A' = \emptyset$. Every point of A is isolated.

- (c) Closure: $\overline{A} = [0, 1]$. First, $A \subseteq [0, 1]$, and $[0, 1]$ is closed, so $\overline{A} \subseteq [0, 1]$. The point 0 is adherent to A : given $r > 0$, choose $n > \max\{1/r, 1\}$, so $1/n \in (-r, r) \cap A$. Now fix $x \in (0, 1]$. By density of $\mathbb{Q}$ in $\mathbb{R}$, for every $r > 0$ there is $q \in \mathbb{Q}$ between $\max\{x - r, 0\}$ and x , and this $q \in (x - r, x + r) \cap A$.

Isolated points: $\emptyset$. Given $x \in A$ and $r > 0$, choose $q \in \mathbb{Q}$ as above. This q lies in $(x - r, x + r) \cap A$ and $q \neq x$.

Limit points: $A' = [0, 1]$. There are no isolated points.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement

Homework 9, Problem 2

Let (X, d) be a metric space, let A be a nonempty subset of X , and let U be an open subset of X .

- (a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.
- (b) Prove that $\overline{U \cap \overline{A}} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)
- (c) Conclude that if $U \cap A = \emptyset$, then $U \cap \overline{A} = \emptyset$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why is the assumption that U is open essential in part (a)?

2. After fixing $x \in U \cap \overline{A}$ and $r > 0$, why do we introduce a second radius $s > 0$?

3. Why is $\tilde{r} = \min\{r, s\}$ the right radius to use?

4. Which closure property turns part (a) into the first inclusion in part (b)?

5. In part (c), why does $\overline{U \cap \overline{A}} = \emptyset$ force $U \cap \overline{A} = \emptyset$?

Part (a): prove $U \cap \overline{A} \subseteq \overline{U \cap A}$.

Fix $x \in \underline{\hspace{2cm}}$. **To show** $x \in \overline{U \cap A}$, **fix** $r > 0$ **and aim to find a point in:**

Use openness of U : since $x \in U$, choose $s > 0$ such that:

Use $x \in \overline{A}$ **with** $\tilde{r} = \min\{r, s\}$ **to choose** $y \in$:

Check the three memberships $y \in A$, $y \in U$, **and** $y \in B_r(x)$:

Part (b): prove equality of closures.

First inclusion from part (a) and monotonicity of closure:

Second inclusion from $A \subseteq \overline{A}$:

Part (c): empty-set conclusion.

Start from $U \cap A = \emptyset$, take closures, use part (b), and then use $B \subseteq \overline{B}$:

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

- (a) Fix $x \in U \cap \overline{A}$. We show $x \in \overline{U \cap A}$, i.e., for any $r > 0$, $B_r(x) \cap U \cap A$ is nonempty. Fix $r > 0$. Since $x \in U$ and U is open, there exists $s > 0$ such that $B_s(x) \subseteq U$. Set $\tilde{r} = \min\{r, s\}$. Since $x \in \overline{A}$, the set $B_{\tilde{r}}(x) \cap A$ is nonempty, so choose $y \in B_{\tilde{r}}(x) \cap A$. Then $y \in A$. Also $y \in B_r(x)$, since $d(y, x) < \tilde{r} \leq r$. Finally $y \in U$, since $d(y, x) < \tilde{r} \leq s$, so $y \in B_s(x) \subseteq U$. Hence $y \in B_r(x) \cap U \cap A$, as desired.
- (b) Recall that $B \subseteq C$ implies $\overline{B} \subseteq \overline{C}$. Applying this to part (a),

$$\overline{U \cap \overline{A}} \subseteq \overline{U \cap A}.$$

On the other hand, $A \subseteq \overline{A}$, so $U \cap A \subseteq U \cap \overline{A}$. Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together,

$$\overline{U \cap \overline{A}} = \overline{U \cap A}.$$

- (c) For any set B , $B \subseteq \overline{B}$. If $U \cap A = \emptyset$, then $\overline{U \cap A} = \emptyset$. By part (b),

$$\overline{U \cap \overline{A}} = \emptyset.$$

Therefore $U \cap \overline{A} = \emptyset$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement

Homework 9, Problem 3

Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y .

- (a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.
- (b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What does a ball in the subspace Y look like compared with a ball in $\mathbb{R}^2$?

2. Why is the line segment $(-1, 1) \times \{0\}$ open in Y ?

3. Why does the point $(0, r/2)$ prove that this set is not open in $\mathbb{R}^2$?

4. What lecture result lets us reduce part (b) to proving that Y is closed in $\mathbb{R}^2$?

5. How does the vertical coordinate $x_2 \neq 0$ give a ball contained in Y^c ?

Part (a): example open in Y , not open in $\mathbb{R}^2$.

Choose $A =$

To prove A is open in Y , fix $x = (x_1, 0) \in A$ and choose $r =$

If $y \in B_r^Y(x)$, show $|y_1| < 1$, hence $y \in A$:

To prove A is not open in $\mathbb{R}^2$, fix $r > 0$ around $(0, 0)$ and find an off-axis point:

Part (b): closedness transfer.

State the lecture corollary being used:

Part (b): closedness transfer, continued.

Prove Y is closed by fixing $x = (x_1, x_2) \in Y^c$, so $x_2 \neq 0$, and choosing radius $r =$

Show $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$ using the inequality involving $|x_2 - y_2|$:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Consider

$$A = \{(x_1, 0) : -1 < x_1 < 1\}.$$

First, A is open in (Y, d_Y) . Fix $x = (x_1, 0) \in A$, and set $r = 1 - |x_1|$. Then $r > 0$, and

$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$

since

$$y \in B_r^Y(x) \implies r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1|$$

and therefore

$$|y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1.$$

Next, A is not open in $(\mathbb{R}^2, d_2)$, because $(0, 0) \in A$ is not an interior point. For any $r > 0$, the point $(0, r/2)$ lies in $B_r^{\mathbb{R}^2}((0, 0))$ but not in A .

(b) By the corollary from Lecture 24, it suffices to show that Y is closed in $(\mathbb{R}^2, d_2)$. Fix $x = (x_1, x_2) \in Y^c$. Then $x_2 \neq 0$. Set $r = |x_2|$. If $y \in B_r^{\mathbb{R}^2}(x)$, then

$$r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2|.$$

Hence

$$|y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0.$$

Thus $y_2 \neq 0$, so $y \in Y^c$. Therefore $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$. Since every point of Y^c is interior to Y^c , Y^c is open, and Y is closed.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement

Homework 9, Problem 4

Let (X, d) be a metric space, $\{x_n\}_{n \geq 1}$ a sequence in X , and $x \in X$. Prove that the sequence $\{x_n\}_{n \geq 1}$ converges to x if and only if for every open set U that contains x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. In the forward direction, how does openness of U connect an arbitrary neighborhood to a ball around x ?

2. What does “for all but finitely many n ” mean in terms of choosing an N ?

3. In the reverse direction, why is $B_r(x)$ the correct open set to test?

4. Why is it enough to prove $x_n \in B_r(x)$ eventually for each $r > 0$?

5. What is the main logical difference between the two directions?

Forward direction: assume $x_n \rightarrow x$.

Fix an open set $U \subseteq X$ with $x \in U$. By openness, choose $r > 0$ such that:

By convergence, choose N such that for all $n \geq N$:

Conclude $x_n \in U$ for all but finitely many n :

Reverse direction: assume eventual membership in every open $U \ni x$.

Fix $r > 0$. Apply the hypothesis to the open set:

Convert “all but finitely many” into an N so that for all $n \geq N$:

Translate $x_n \in B_r(x)$ into the metric convergence inequality:

Conclude because $r > 0$ was arbitrary:

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Suppose $x_n \rightarrow x$ in (X, d) as $n \rightarrow \infty$. Fix an open set $U \subseteq X$ such that $x \in U$. Since U is open, there exists $r > 0$ such that $B_r(x) \subseteq U$. By definition of convergence, there exists N such that $d(x_n, x) < r$ for all $n \geq N$. Therefore $x_n \in B_r(x) \subseteq U$ for all $n \geq N$.

$\Leftarrow$ Suppose that for every open set U containing x , we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$. Fix $r > 0$. Apply the assumption to the open set $B_r(x)$. Then $x_n \in B_r(x)$ for all but finitely many $n \in \mathbb{N}$. Let N be larger than the largest index for which $x_n \notin B_r(x)$. Then for every $n \geq N$, $x_n \in B_r(x)$, and hence $d(x_n, x) < r$. Since $r > 0$ was arbitrary, $\{x_n\}_{n \geq 1}$ converges to x in (X, d) .

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement

Homework 9, Problem 5

Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

- (a) Prove that $|x_1^n - x_1| \leq d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.
- (b) Prove that the sequence $\{x^n\}_{n \geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n \geq 1}$ and $\{x_2^n\}_{n \geq 1}$ of real numbers both converge.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. Why does the Euclidean distance $d_2(x^n, x)$ control each coordinate difference?

2. In the forward direction, what is squeezed between 0 and $d_2(x^n, x)$?

3. Why do the coordinate sequences converge to the coordinates of the same point x ?

4. In the reverse direction, how do the real limit laws imply $d_2(x^n, x) \rightarrow 0$?

5. What would change if the problem were in $\mathbb{R}^3$ or $\mathbb{R}^m$?

Part (a): coordinate estimate.

Write the formula for $d_2(x^n, x)$:

Use $|x_2^n - x_2|^2 \geq 0$ to get the desired inequality:

Part (b), forward direction.

Assume $x^n \rightarrow x = (x_1, x_2)$ in $(\mathbb{R}^2, d_2)$. Then $d_2(x^n, x) \rightarrow$

Use part (a) and squeeze to prove $x_1^n \rightarrow x_1$:

Repeat the same argument for the second coordinate:

Part (b), reverse direction.

Assume $x_1^n \rightarrow x_1$ and $x_2^n \rightarrow x_2$. Set $x =$

Use the formula for $d_2(x^n, x)$ and the limit laws to conclude:

Use this page for part (a) and the forward direction of part (b).

Use this page for the reverse direction of part (b).

Official Solution (Professor)

(a) Recall that

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2}.$$

Since $|x_2^n - x_2|^2 \geq 0$,

$$d_2(x^n, x) \geq \left(|x_1^n - x_1|^2 + 0\right)^{1/2} \geq |x_1^n - x_1|.$$

(b) $\Rightarrow$ Suppose $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. Then

$$d_2(x^n, x) \rightarrow 0.$$

By part (a),

$$0 \leq |x_1^n - x_1| \leq d_2(x^n, x).$$

By the squeeze theorem, $|x_1^n - x_1| \rightarrow 0$, so $\{x_1^n\}_{n \geq 1}$ converges to x_1 . Similarly,

$$0 \leq |x_2^n - x_2| \leq d_2(x^n, x),$$

so $\{x_2^n\}_{n \geq 1}$ converges to x_2 .

$\Leftarrow$ Suppose $\{x_1^n\}_{n \geq 1}$ converges to x_1 and $\{x_2^n\}_{n \geq 1}$ converges to x_2 . Then

$$|x_1^n - x_1| \rightarrow 0 \quad \text{and} \quad |x_2^n - x_2| \rightarrow 0.$$

By the limit laws,

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2} \rightarrow 0.$$

Thus $\{x^n\}_{n \geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement

Homework 9, Problem 6

Let (X, d) be a metric space and $\{x_n\}_{n \geq 1}$ a sequence in X . Prove that if $\{x_n\}_{n \geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n \geq 1}$ of real numbers converges to $d(x, y)$.

Topic: _____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

1. What real-valued sequence are we trying to prove converges?

2. Why should the triangle inequality compare $d(x_n, y)$ and $d(x, y)$?

3. How do two applications of the triangle inequality produce an absolute value estimate?

4. What quantity goes to 0, and why does it squeeze the distance error?

5. How is this problem a sequence version of continuity of the map $z \mapsto d(z, y)$?

Setup.

Fix $y \in X$. Since $x_n \rightarrow x$, we know $d(x_n, x) \rightarrow$

Triangle inequality estimates.

First apply triangle inequality to bound $d(x_n, y)$ in terms of $d(x_n, x)$ and $d(x, y)$:

Rearrange to get $d(x_n, y) - d(x, y) \leq$

Apply triangle inequality again to bound $d(x, y)$ in terms of $d(x, x_n)$ and $d(x_n, y)$:

Rearrange to get $d(x, y) - d(x_n, y) \leq$

Finish.

Combine the two inequalities into an absolute value estimate:

Use squeeze theorem to conclude:

Use this page for the complete proof.

Official Solution (Professor)

Fix $y \in X$. By the triangle inequality,

$$\begin{aligned}d(x_n, y) &\leq d(x_n, x) + d(x, y) &\implies d(x_n, y) - d(x, y) &\leq d(x_n, x), \\d(x, y) &\leq d(x, x_n) + d(x_n, y) &\implies d(x, y) - d(x_n, y) &\leq d(x_n, x).\end{aligned}$$

Together,

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

Since $\{x_n\}_{n \geq 1}$ converges to x in (X, d) , the right-hand side converges to 0. Therefore, by the squeeze theorem,

$$|d(x_n, y) - d(x, y)| \rightarrow 0,$$

and thus $d(x_n, y)$ converges to $d(x, y)$.

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:
